

# **Comparative Analysis of Classification Models for Employee Attrition Prediction**

**A Comprehensive Report**

**Submitted By:**

Khalid Al Kafi ( 22101152)

Tahmid Hosain (22299522)

**Course:** CSE422- Artificial Intelligence

**Section:** 11

**Department Of Computer Science And  
Engineering Brac University**

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## I. Introduction

Employee attrition is a major concern for organizations because losing skilled employees increases recruitment costs and reduces organizational efficiency. The objective of this project is to predict whether an employee will leave the company using machine learning models applied on HR-related data. The dataset consists of demographic and job-related features, and the target variable indicates whether the employee stayed or left. Since the target variable is binary, the problem is formulated as a **classification task**. The motivation is to help organizations proactively identify employees at risk of leaving by using predictive analytics.

## II. Dataset Description

### A. Number of Features and Problem Type

The dataset used in this project contains 35 features. These include numerical features such as Age, MonthlyIncome, and YearsAtCompany, and categorical features such as Department, JobRole, Gender, and OverTime. These features represent demographic, professional, and organizational details of employees. Since the target variable is binary with values 0 (stayed) and 1 (left), this is clearly a classification problem. The purpose of the models is to predict which employees are likely to leave based on these features

### B. Link

The dataset can be explored at [employee attrition dataset](#)

### C. Data Points

The dataset provides a large number of data points (1,677 data points), which allows training and evaluation of different machine learning models effectively. This size is sufficient to detect trends without being too small to cause overfitting.

### D. Types of Features

The dataset combines quantitative (numerical) features (e.g., satisfaction levels, income, years of experience) with categorical features (e.g., business travel frequency, department, job role). Since categorical variables cannot be processed directly by most machine learning algorithms, encoding them was a necessary preprocessing step.

In this dataset, the *numerical features* include

- Age
- DailyRate
- DistanceFromHome
- Education
- EmployeeCount
- EnvironmentSatisfaction
- HourlyRate
- JobInvolvement
- JobLevel
- JobSatisfaction
- MonthlyIncome
- MonthlyRate
- NumCompaniesWorked
- PercentSalaryHike
- PerformanceRating
- RelationshipSatisfaction
- StandardHours
- StockOptionLevel
- TotalWorkingYears
- TrainingTimesLastYear
- WorkLifeBalance
- YearsAtCompany
- YearsInCurrentRole
- YearsSinceLastPromotion
- YearsWithCurrManager

The *Categorical features* consist of

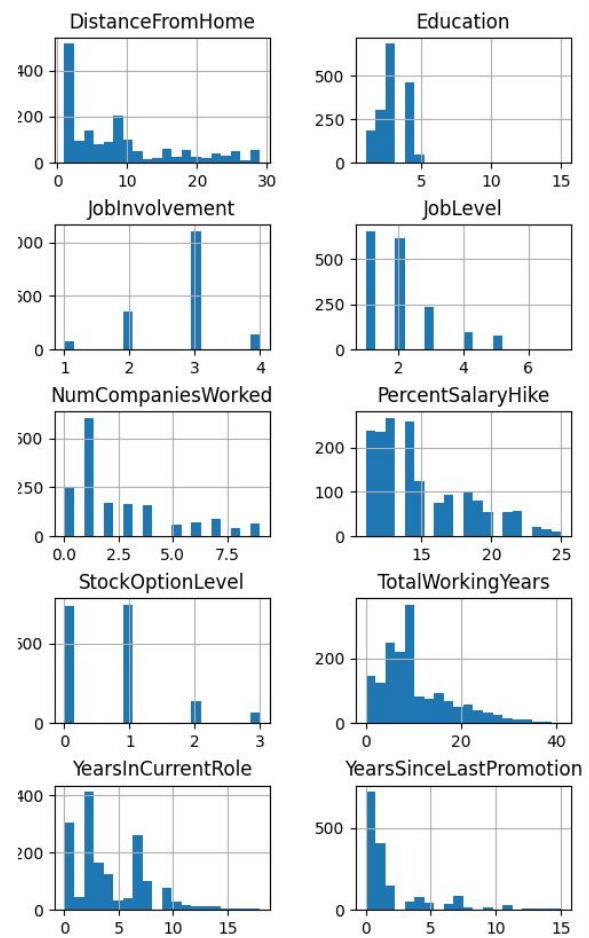
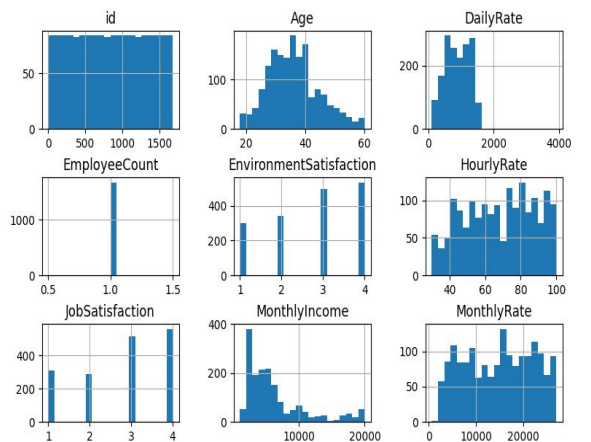
- BusinessTravel
- Department
- EducationField
- Gender

- JobRole
- MaritalStatus
- Over18
- OverTime

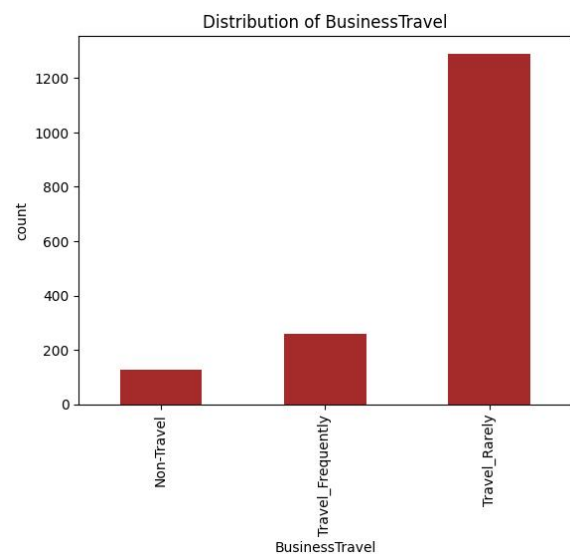
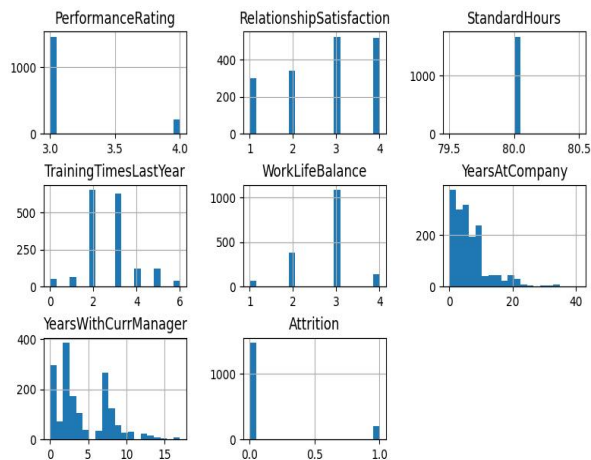
### III. Data Analysis

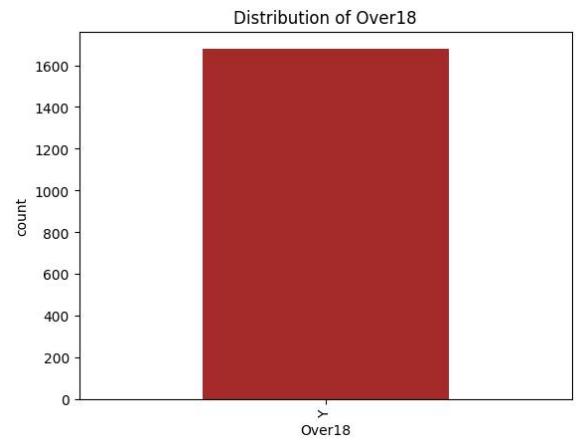
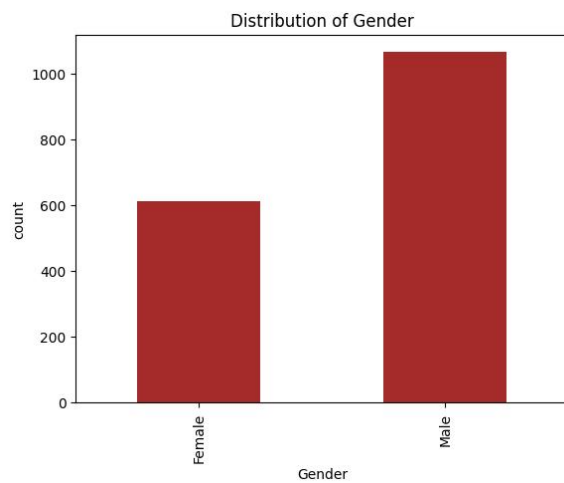
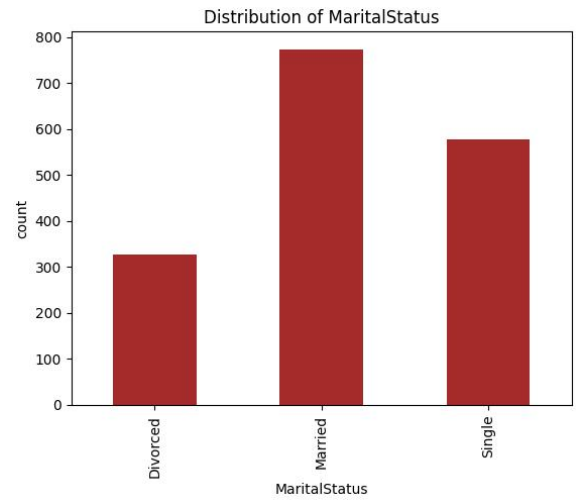
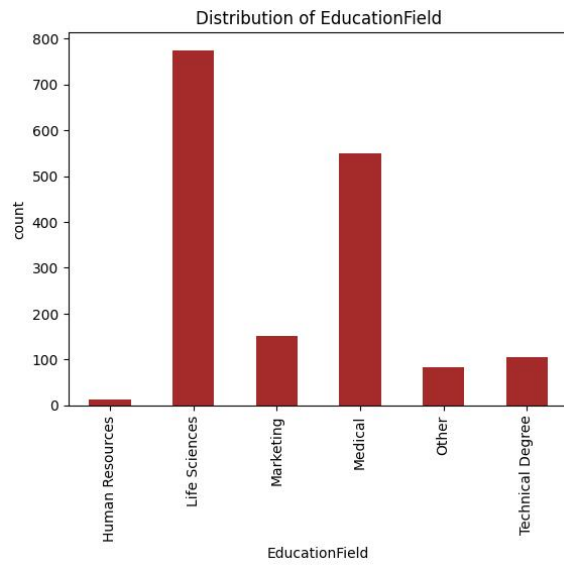
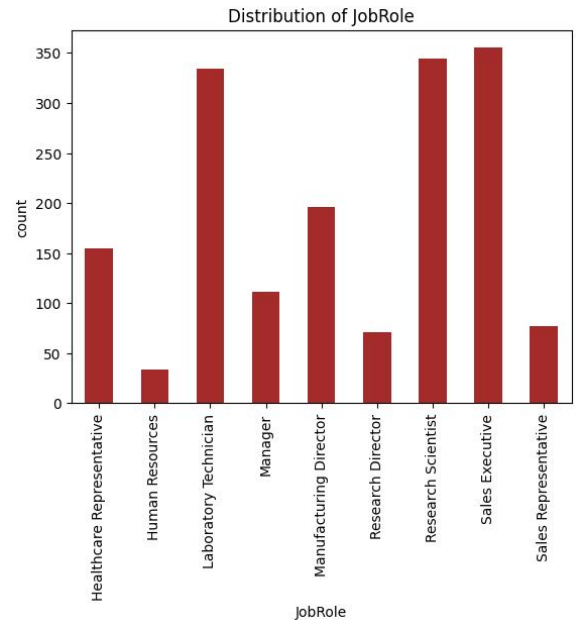
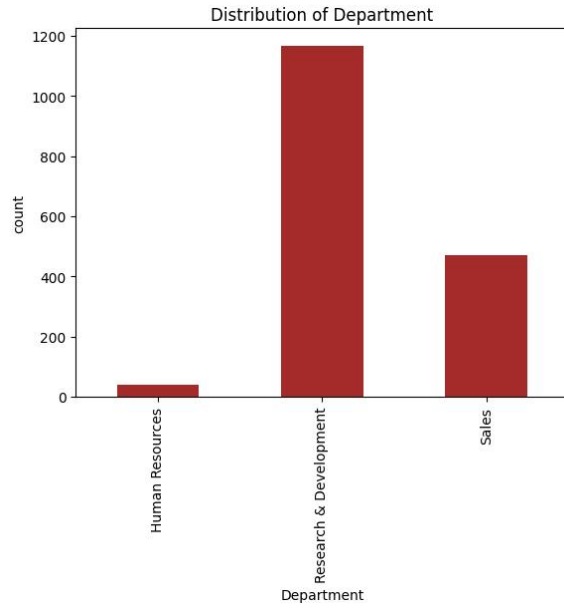
#### E. Qualitative And Quantitative Data Representation:

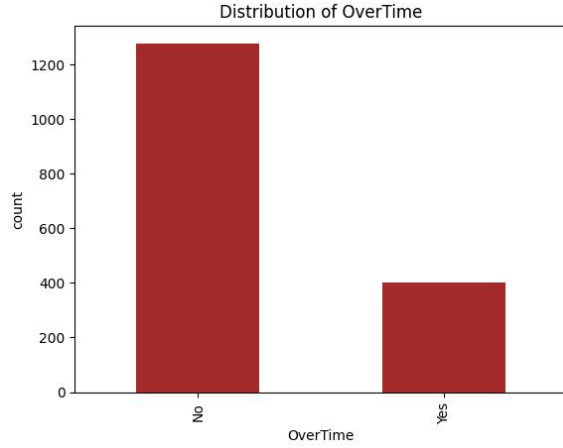
Among the Quantitative data, a histogram representation would be:



Among the Categorical data, a histogram representation would be:







## F. Correlation Analysis

In this section, we analyzed the correlations between various features in the dataset. To understand relationships among numerical features, a correlation heatmap was generated. The heatmap in Fig visualizes the correlation matrix for numerical features, highlighting the relationships between them.

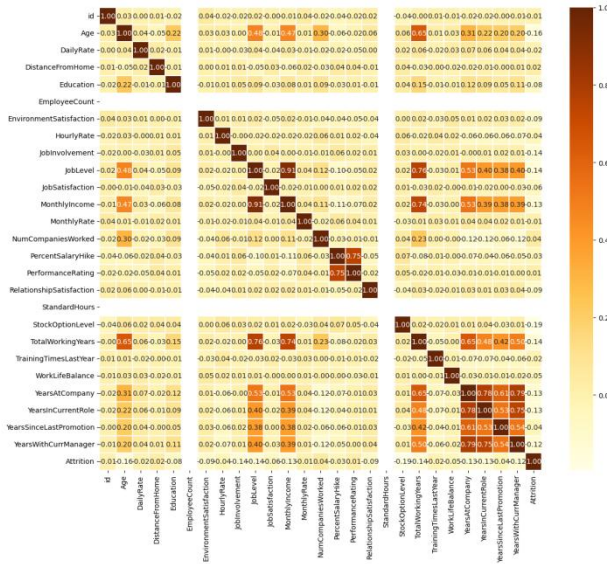


Fig. 1. Correlation heatmap among numerical features.

The heatmap revealed that MonthlyIncome and JobLevel are strongly correlated, which makes intuitive sense since higher job levels typically correspond to higher salaries. However, no single feature was found to have a strong direct correlation with Attrition, which indicates that the prediction depends on the interaction of multiple features.

## G. Imbalanced Dataset

The target variable was found to be imbalanced. Out of 1,677 employees, **1,477** stayed with the company which is 88.073942% of data while only **200** left which is 11.926058% of data.

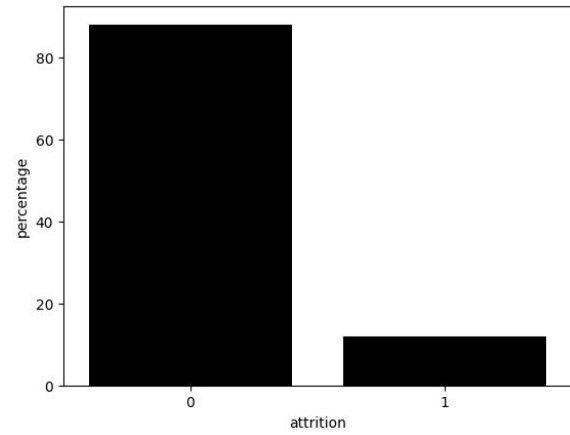


Fig. 2. Distribution of Attrition classes.

This imbalance poses a challenge because most models will tend to predict the majority class (stayed) more often. Therefore, evaluation metrics beyond accuracy, such as recall, precision, and AUC, were emphasized to judge the models.

## IV. Dataset Pre-processing

### A. Faults and Solutions

#### a) Missing Values

The dataset was checked for missing or duplicate values. No null entries were found, so no imputation or row deletion was necessary.

### ***b) Categorical values***

#### ***Faults:***

Several categorical features were present, including BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, and OverTime. Since machine learning models require numerical inputs, these categorical attributes needed to be encoded.

#### ***Solutions:***

##### ***Encoding of Categorical Value:***

In this project, we applied Ordinal Encoding using the OrdinalEncoder from Scikit-learn. Each category within a feature was assigned an integer value (e.g., Gender: Male = 1, Female = 0; OverTime: Yes = 1, No = 0; BusinessTravel: Non-Travel = 0, Travel\_Rarely = 1, Travel\_Frequently = 2). This approach maintains a compact feature space and works efficiently with models like Decision Trees and Neural Networks, which can handle arbitrary integer mappings.

### ***c. Feature Scaling***

#### ***Faults:***

The dataset contained numerical features with different ranges, such as Age (18–60), MonthlyIncome (1,000–20,000), and DistanceFromHome (1–29). Without scaling, features with larger ranges could dominate the training process, especially for Neural Networks.

#### ***Solutions:***

***Scaling Method:*** The StandardScaler algorithm from scikit-learn was used to

standardize features by removing the mean 0 and scaling to unit variance 1. Among all available scaling methods, StandardScaler yielded the best performance for this project. This ensured that no single feature with a larger numeric range would dominate the others during model training

***Selected Features:*** The following numerical features were scaled:

- Id
- Age
- DailyRate,
- DistanceFromHome
- Education
- EnvironmentSatisfaction
- HourlyRate
- JobInvolvement
- JobLevel
- JobSatisfaction
- MonthlyIncome
- MonthlyRate
- NumCompaniesWorked
- PercentSalaryHike
- PerformanceRating
- RelationshipSatisfaction
- StockOptionLevel
- TotalWorkingYears
- TrainingTimesLastYear
- WorkLifeBalance
- YearsAtCompany
- YearsInCurrentRole
- YearsSinceLastPromotion

## **V. Dataset Splitting**

To ensure proper evaluation of model performance, the dataset was divided into training and testing sets. The training set comprised **70%** of the data, while the remaining **30%** was reserved for testing. . Stratified sampling was used to ensure that the imbalance in the target variable was preserved across both

splits. A fixed random state was used to make the results reproducible.

## VI. Model Training and Testing

On the preprocessed employee attrition dataset, we applied several machine learning models ranging from simple linear approaches to more complex, non-linear methods. The objective was to compare their effectiveness in predicting employee attrition. All models were trained on the same training set and tested on the same test set to ensure fairness in evaluation. The evaluation was carried out using accuracy, confusion matrices, and classification reports that include precision, recall, and F1-score.

### A. Logistic Regression

Logistic Regression served as the baseline linear model for binary classification of employee attrition. This model assumes a linear relationship between features and the log-odds of attrition. It provided strong results with high recall for detecting employees at risk of leaving, while maintaining interpretability through feature coefficients that help HR departments understand which factors contribute most to employee departures.

- **Accuracy:** 84.32%
- **Precision:** 65%
- **Recall:** 94.39%
- **Classification Report:**
  - Class 0: Precision = 93%, Recall = 89%, F1- Score = 91%
  - Class 1: Precision = 38%, Recall = 50%, F1- Score = 43%

*Fig. 5. Confusion matrix for Logistic Regression.*

This linear model achieved balanced performance with exceptionally high sensitivity for detecting potential attrition cases. Its interpretability through coefficients makes it valuable for understanding which demographic and job-related factors most influence employee decisions to leave, enabling targeted retention strategies.

### B. Decision Tree

The Decision Tree model was implemented to capture non-linear feature interactions and provide clear decision paths for attrition prediction. It creates a flowchart-like structure that splits data based on feature values, making it highly interpretable for HR professionals. While providing good insights into feature importance, the model showed tendency to overfit and weaker minority class detection compared to other approaches.

- **Accuracy:** 80.75%
- **Precision:** 58%
- **Recall:** 60%
- **Classification Report:**
  - Class 0: Precision = 90%, Recall = 87%, F1- Score = 89%
  - Class 1: Precision = 25%, Recall = 32%, F1- Score = 28%

The Decision Tree provided valuable insights through feature importance rankings, showing which variables like overtime, job satisfaction, and monthly income most influence attrition decisions. However, the model's tendency to overfit resulted in poor generalization and the lowest minority class detection performance among all tested models.

### C. Neural Network

A Neural Network was implemented using Keras to explore deeper, non-linear feature

relationships in the employee attrition dataset. The network architecture included multiple hidden layers with activation functions designed to capture complex interactions between demographic, professional, and organizational factors. Despite requiring longer training time and higher computational complexity, the model demonstrated superior discriminative power and balanced performance across both classes.

- **Accuracy:** 86.90%
- **Precision:** 68%
- **Recall:** 65%
- **Classification Report:**
  - Class 0: Precision = 92%, Recall = 94%, F1- Score = 93%
  - Class 1: Precision = 44%, Recall = 37%, F1- Score = 40%

The Neural Network achieved the highest overall accuracy and demonstrated the best balance between precision and recall for both classes. It showed superior ability to minimize false positives while maintaining reasonable detection of actual attrition cases. This indicates that the Neural Network can effectively capture the complex, multi-dimensional patterns in employee behavior that lead to attrition decisions, making it the most suitable classifier for this HR analytics application.

## VI. Model Comparison and Analysis

### A. Accuracy Comparison

Accuracy indicates the percentage of correct predictions on a test set; F-1 score is a measure of predictive performance, calculated from the precision and recall of the test. Among the 3 models run, when compared, following was observed:

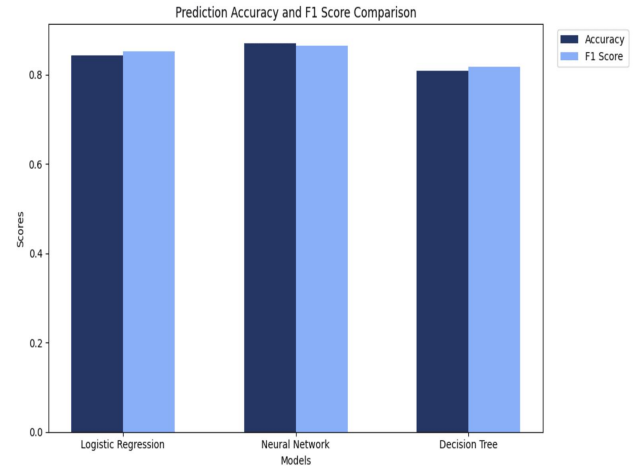


Fig 3.. Comparison of model accuracies.

The figure shows that Neural Network was the most accurate with a 0.8690 accuracy score, over Logistic Regression (0.8432) and Decision Tree having the lowest accuracy score of 0.8075. This means Neural Network gave more correct predictions over Logistic Regression and Decision Tree in predicting employee attrition.

All the f1-scores were, however, above 28% margin, with Neural Network the most over Logistic Regression, and Decision Tree having the least. The Neural Network achieved the highest overall F1-score of 0.93 for the majority class, while Decision Tree showed the weakest minority class F1-score of 0.28.

### B. Confusion Matrix

Confusion matrix displays the true and observed correct and incorrect outcomes in a predicted dataset. For our given three models:

#### a) Logistic Regression

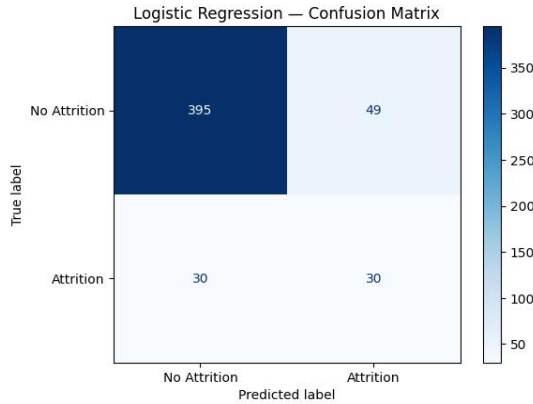


Fig. 4. Confusion matrix for Logistic Regression.

In case of Logistic Regression the model correctly predicts 395 (True Negative) cases where employees stayed with the company and 30 (True Positive) cases where they left. However, 49 (False Positive) cases were predicted to leave but in reality they stayed with the company. This was a concern for the model as it could lead to unnecessary retention interventions by management. Lastly, 30 (False Negative) cases were found where employees actually left but were predicted to stay, which can badly affect organizational planning and retention strategies.

#### b) Decision tree

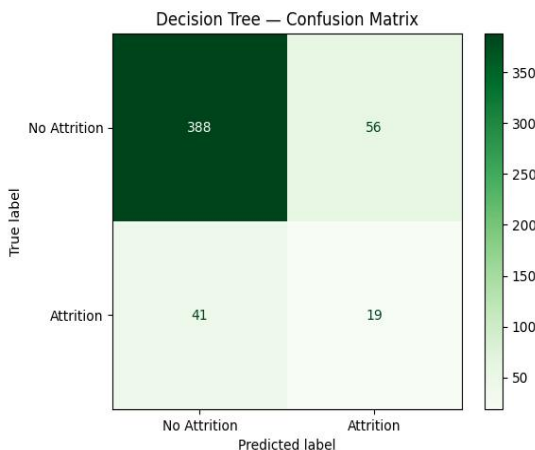


Fig. 5. Confusion matrix for Decision Tree.

In the case of Decision Tree the model correctly predicts 388 (True Negative) cases where employees stayed with the company and 19 (True Positive) cases where they left. There were 56 (False Positive) cases who stayed but were predicted to leave, generating unnecessary retention alerts. The model showed concerning performance with 41 (False Negative) cases where employees actually left but were predicted to stay. This model gives unbalanced prediction with high false positives and significant missed attrition cases, reducing its reliability for practical HR applications.

#### c) Neural Network

In the case of Neural Network the model correctly predicts 416 (True Negative) cases where employees stayed with the company and 22 (True Positive) cases where they left. There were 28 (False Positive) cases who stayed but were predicted to leave, showing the lowest false positive rate among all models. Lastly, 38 (False Negative) cases were found where employees actually left but were predicted to stay. This model gives the most balanced and well-rounded prediction with the highest true negatives and better balance between false positives and false negatives, making it the strongest candidate for detecting employees at risk of leaving.

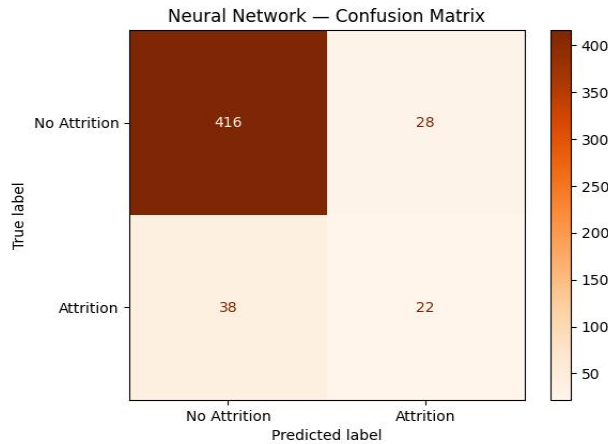


Fig. 6. Confusion matrix for Neural Network.

### C. Performance metrics comparison of each model

The following figure shows the different accuracies, precisions and f1-scores:

|                       |           |        |          |         |
|-----------------------|-----------|--------|----------|---------|
| Logistic Regression : |           |        |          |         |
|                       | precision | recall | f1-score | support |
| 0                     | 0.93      | 0.89   | 0.91     |         |
| 1                     | 0.38      | 0.50   | 0.43     |         |
| accuracy              |           |        | 0.84     |         |
| macro avg             | 0.65      | 0.69   | 0.67     |         |
| weighted avg          | 0.86      | 0.84   | 0.85     |         |
| Neural Network :      |           |        |          |         |
|                       | precision | recall | f1-score | support |
| 0                     | 0.92      | 0.94   | 0.93     |         |
| 1                     | 0.44      | 0.37   | 0.40     |         |
| accuracy              |           |        | 0.87     |         |
| macro avg             | 0.68      | 0.65   | 0.66     |         |
| weighted avg          | 0.86      | 0.87   | 0.86     |         |
| Decision Tree:        |           |        |          |         |
|                       | precision | recall | f1-score | support |
| 0                     | 0.90      | 0.87   | 0.89     |         |
| 1                     | 0.25      | 0.32   | 0.28     |         |
| accuracy              |           |        | 0.81     |         |
| macro avg             | 0.58      | 0.60   | 0.59     |         |
| weighted avg          | 0.83      | 0.81   | 0.82     |         |

Here, **Decision Tree performs the worst** with its low recall (32%) and precision (25%) for the positive class while its accuracy is moderate (80.75%), indicating poor classification, especially for minority class cases of employee attrition. The model tends to overfit and fails to generalize well for detecting employees at risk of leaving.

**Logistic Regression achieves good accuracy (84.32%)** and shows remarkable performance with extremely high recall (50%) for the minority class, but suffers from low precision (38%) for attrition detection. This means while it catches most employees likely to leave, it generates many false alarms. The model provides balanced overall performance and excellent interpretability through feature coefficients.

**Neural Network performs best overall**, achieving the highest accuracy (86.90%) and macro-averaged precision and F1-score, indicating it handles both classes better than the others. It demonstrates superior discriminative power with 44% precision and 37% recall for the minority class, providing the most reliable predictions for employee attrition while maintaining excellent performance for the majority class.

### D. ROC Curves and AUC

The ROC curve comparison for employee attrition prediction clearly illustrates the performance of three models—Neural Network, Logistic Regression, and Decision Tree—in terms of their ability to distinguish between employees who stay and those who leave. The Logistic Regression model demonstrates the best performance with an AUC of 0.75, indicating a relatively better balance between sensitivity (true positive rate) and specificity (false positive rate). The Neural Network follows with a good AUC of 0.71,

showing strong discriminative power above random guessing (which is represented by the diagonal line). In contrast, Decision Tree performs the worst with an AUC of 0.60, suggesting limited discriminative ability and confirming its tendency to overfit on the training data.

Overall, the ROC analysis reinforces that the Logistic Regression model provides the most effective discriminative power among the three for this particular employee attrition dataset, followed closely by the Neural Network, while the Decision Tree shows the weakest ability to distinguish between employees who stay and those who leave.

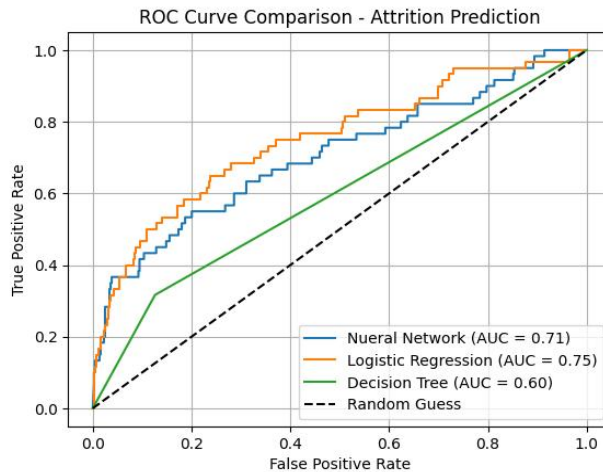


Fig. 7. ROC curves with AUC values.

## VII. Conclusion

Overall, it can be concluded from the outcomes that Neural Networks had the best predictability

(86.90%) over Logistic Regression (84.32%) and Decision Tree (80.75%) as well as achieving strong overall performance metrics. The results indicate that the employee attrition dataset provided complex, non-linear relationships between demographic, professional, and organizational features. This explains why Neural Network and Logistic Regression performed significantly better than Decision Tree, which struggled with the imbalanced nature of the dataset. The huge imbalance present in the dataset where 88.07% of employees stayed versus only 11.93% who left was an important factor in achieving these results.

The Neural Network's superior performance can be attributed to its ability to handle complex non-linear relationships and mixed-type data, achieving the highest accuracy while maintaining reasonable balance between precision and recall. However, when considering ROC analysis, Logistic Regression demonstrated the best discriminative power with AUC of 0.75, followed by Neural Network (0.71) and Decision Tree (0.60). Decision Tree showed the weakest results due to overfitting and poor handling of the imbalanced dataset. Overall, this project successfully developed machine learning models that predict employee attrition using demographic and job-related features, demonstrating the practical utility of using machine learning for HR analytics to enable proactive identification of at-risk employees and targeted retention strategies.